

Part 5 — Market linkage: the channels, and what "one tap" actually means

Q5.1 — "One tap to Amazon, Flipkart, Meesho, GeM and ONDC. Which of those have you actually published a listing to?"

Think. One. Say "one" first. Then give the tier table, because the tiering is the answer and it is stronger than the claim it replaces.

Answer. One — our own marketplace. Everything else is mapped, rendered or connected, and I will tell you exactly which is which.

Channel	Tier	State today	Blocker
Our marketplace	A	Live. routers/publish.py persists it, routers/marketplace.py serves it	—
ONDC	A	Catalog mapped , returns <code>dry_run</code>	No becn-onix subscriber id issued yet
GeM	C	<code>.xlsx</code> renderer built, <code>openpyxl</code>	Zero real category templates in the repo — <code>gem_templates/</code> holds a README and nothing else
Amazon	B	OAuth real, payload real, PUT not implemented	<code>TODO(phase 8)</code> in <code>channels/amazon.py</code>
Flipkart	B	Same shape, 51 lines	Same
Meesho	D	Copy-paste block + guided browser	No public API exists
WhatsApp	D	Image + caption	—

Our own registry says it, unprompted, in a docstring (`web/api/channels/registry.py`):

"This said 'two live listings ... and it is real' while ONDC was returning a fabricated `ondc:{id}`. Put the two back when the push is real. The number here is the demo claim, and a demo claim that outruns the code is how you lose the room."

The Amazon adapter refuses to lie for the same reason, and I would read this line aloud if asked: *"An artisan who has done the work of connecting their Seller Central account has earned an accurate answer about what we then did*

with it." It returns `dry_run` with the message "SP-API listings PUT not implemented — nothing was sent to Amazon."

Why the tiering is the better pitch. A flat "one-click everywhere" claim does not survive one industry judge. **The tiering proves we know which Indian platforms have APIs and which do not** — and that knowledge is the actual product. Anyone can call an API. Knowing that GeM has none, that Meesho is partner-gated, that Myntra requires a **trademark certificate a weaver cannot possibly have** — that is the work.

Dropped deliberately, and this matters: Myntra, Nykaa, Ajio. Myntra onboarding requires GST, PAN, Aadhaar, **a trademark certificate or brand authorisation letter**, address proof and incorporation documents. **A weaver has no trademark.** If an industry judge is in the room, claiming Myntra integration is instant credibility loss. They are off every slide.

Q5.2 — "GeM has no seller API. So how is uploading a spreadsheet 'AI-driven market linkage'?"

Think. This is where our best story is, and it is counter-intuitive enough that it needs the number up front.

Answer. GeM has 10,700+ product categories.

Choosing the wrong one is the single most common listing failure — errors in category selection and attribute accuracy cause delayed approvals, repeated rejections and lost opportunities. **An artisan cannot navigate 10,700 categories. Neither can most educated sellers: an entire consultancy industry exists purely to fill GeM catalogues correctly.**

A vision model maps a photograph of a Sambalpuri saree to the right GeM category. NLP fills the attribute schema. We look up the HSN code.

*That is a consultancy industry, automated, free, in the artisan's own language. Background removal is a commodity. **This is not.** This is what "AI-driven market linkage" actually means.*

And the Excel path is the official one, not a workaround. Per GeM's own Catalogue Management guidance: sellers may use the bulk upload facility with the category-specific Excel format, and once filled accurately and saved, **the item is automatically listed on GeM 3.0.** So our job is precisely stated: **generate a perfect, category-correct GeM Excel from a photograph and a voice note.**

The blocker, and it is in the adapter's own docstring as [warning] BLOCKING: the real category templates are `docs/Master-Technical-Reference.md` §18 items 1 and 2 and **are not in the repo.** `_load_template` reads a JSON descriptor per category "so that when the real templates arrive, adding one is dropping in a file — never editing this adapter." The architecture is right and the data is missing. **Phase 7 cannot start without it**, and it is the single highest-priority unassigned task in the project.

The GeM facts that make this channel the right primary, and which most teams will not know:

Fact	Detail
Caution money	Artisans and weavers FULLY EXEMPT
Transaction charges	Zero up to ₹10 lakh order value
Annual Milestone Charge	One-time ₹10,000 only after ₹20 lakh Seller Merchandise Value in a FY
Registration	Free, online, paperless, 1–3 working days
MSE quota	Minimum 25% of government procurement reserved for MSMEs — guaranteed demand
Already on GeM	~1.5 lakh weavers and handloom entities
Top rejection cause	Name mismatch across Aadhaar / PAN / GST / bank

*An artisan will never cross ₹20 lakh SMV. GeM is completely free for our users. No caution money, no transaction charges, no milestone charge, free registration. **The only remaining barrier is knowing how** — and that is exactly the barrier we remove.*

Still unverified and we say so: Vendor Assessment (~₹11,200 + GST) applies to manufacturers/OEMs, and **an artisan is an OEM, not a reseller**. Exemption route exists; **Vendor Validation is still required**. We need it in writing from GeM and we do not have it.

Q5.3 — "The top GeM rejection is a name mismatch. What do you do about it?"

Think. Small feature, disproportionate value, and it demonstrates that we researched failure modes rather than happy paths.

Today. Before an artisan starts *any* government signup, /channels/gem/setup asks them to **say their name three times** — as printed on the PAN, on the bank passbook, on the Aadhaar card. We compare the three transcripts.

Twenty seconds of voice, before they spend three days on a registration that will be rejected. It costs us almost nothing and it is the single highest-value thing on that screen.

And it is privacy-clean by construction: we store the **comparison result**, never the names-as-documents. `docs/Application-Architecture.md §6.1`.

Q5.4 — "You said artisans need no GST. That is a strong claim. On what authority?"

Think. Cite the notification number. This is the one place a specific legal citation wins the room outright, because almost nobody else will know it exists.

Answer. Notification 34/2023-Central Tax, effective 1 October 2023.

Persons supplying goods **through an e-commerce operator** are **exempt from mandatory GST registration** where:

- (a) supplies are within a single State/UT
- (b) no inter-state supply
- (c) they have a PAN
- (d) the PAN is declared on the GST portal with place of business and State/UT
- (e) they obtain an **enrolment number** on the common portal after PAN validation — and cannot supply through an ECO before that

Applicable where aggregate turnover in the previous and current FY is below the State/UT registration threshold. **GSTN has built the enrolment functionality.**

So we built the GST Enrolment Wizard (/wizard/gst, route 20). Three spoken questions — "*Kis rajya me hain? Doosre rajya me bhejenge? Saal me kitna kaam?*" — routing to one of three outcomes:

Situation	Path
Intra-state only, below threshold	No GST needed — enrolment number only. The wizard walks them through
Inter-state intent	Full GST registration required — no exemption
Above threshold	Full registration

*This feature **literally, legally** addresses "drastically lower the barrier to entry," and almost no other team will know this notification exists.*

The reciprocal obligation, which we state rather than hide. Because we become the ONDC Marketplace Seller Node, **we** are the e-commerce operator. Per **CBIC Circular 194/06/2023**, where the supplier-side ECO is not itself the supplier, **it carries the TCS liability** — so TCS collection and **monthly GSTR-8 filings are ours, not the artisan's**.

That is a real, recurring compliance obligation on us. It is the price of Tier A and it is worth it — because it is the only mechanism by which a weaver with a phone and a PAN card can be selling on a national network with no registration of their own. Cost of that compliance is a line item in the costing document, not a footnote.

Q5.5 — "Explain the ONDC claim. Why does an artisan need nothing at all?"

Think. This is our strongest architectural decision. State the mechanism, then the consequence.

Today — the posture, decided and recorded (docs/decisions.md): **we become the ONDC Marketplace Seller Node.** An MSN does not hold its own inventory; it offers other sellers' goods. That is an exact fit. **We register once. Every artisan is a sub-seller under our node.**

Consequences, and each one is a barrier deleted:

- No ONDC registration for the artisan
- No DigiReady certification
- **No GST** — Notification 34/2023 applies because we are the ECO
- One tap → visible across **every ONDC buyer app simultaneously** (Paytm, Magicpin, Mystore, ...). **One integration, many buyer apps. Amazon offers nothing comparable**

| *A weaver with a phone and a PAN card goes from zero to sellable on a national network in one tap.*

What it costs technically. Self-signed certificates with **separate key pairs for signing and encryption**, separate certs for buyer node (BAP) and seller node (BPP), registration on the registry portal with Subscriber ID, country, cities, domain and type; then the subscription workflow and registrar approval. After that we hold `subscriber_id`, `signing_public_key`, `encryption_public_key`, used in every request.

State today: dry_run. The catalog is mapped; **nothing is signed or pushed**, because the subscriber id has not been issued. There is a staging environment and it is the most demoable of all our channels — that is phase 6 and it is the phase that makes the USP real.

Breaks when. Registrar approval is an external dependency on someone else's timeline. If it does not land, Tier A collapses to one channel — our own marketplace — and the "zero paperwork" claim loses its national reach. That is the largest single external risk in the project.

Q5.6 — "You are filling forms on government websites with an injected script. Is that not automation of a portal you do not own?"

Think. They are probing for a Play Store / terms-of-service problem. There is one nearby, we already rejected it, and saying so is the answer.

What we rejected, explicitly and by name (CLAUDE.md, rejected list): an **AccessibilityService overlay on top of the real Amazon/Flipkart apps**. Google Play permits that API only for genuine disability tools; **enforcement tightened 28 January 2026, and Android 17 blocks non-accessibility apps from the API outright**. Misuse means app suspension and developer-account termination. **On a government problem statement that is disqualifying. The idea is dead.**

What we built instead — our WebView, their site. The artisan taps *"GeM par register karein"*, an in-app browser opens gem.gov.in in a container we control, and a bar pins to the bottom that speaks each instruction and puts the exact value on the clipboard. The artisan self-reports done.

Autofill is a layer on top of that, not a replacement. It is legitimate because it is **the artisan's own browser session, on the artisan's own account, filling the artisan's own data, at the artisan's explicit request** — materially different from automating a third party's app from outside.

Four non-negotiable rules, and rule 3 is the one that answers the question:

1. **Selectors live on the server** — versioned JSON, fetched at runtime, never compiled into the app. A DOM change is a config push we ship in an hour.
2. **Every step degrades, never fails.** Selector misses → that step falls back to guided-paste, speaks the instruction, and carries on. The artisan sees a slower step, never a broken screen.
3. **We never automate submit, login, OTP, payment or CAPTCHA. We fill fields. The artisan presses the button.** A human stays in the loop on every consequential action.
4. **Health telemetry per selector.** Match-rate per step per pack version; below threshold → alert, and the channel **auto-reverts to guided-paste** until a new pack ships.

Ships off. Per-channel flag. Turns on only after its selector pack holds >95% match for a week. Rollout order is cheapest DOM first: GeM registration → GST enrolment → Meesho Supplier Hub. Never Amazon or Flipkart listing forms, because Tier B already has a real API.

The honest framing for judges, verbatim from docs/Application-Architecture.md §6.4:

"Tier A and B are one tap through APIs we control. Tier C and D are one tap plus a confirm, through an assisted browser that fills the artisan's own forms on the artisan's own account — and degrades to spoken step-by-step guidance the moment a page changes."

Breaks when. A portal redesign. That is *when*, not *if* — which is exactly why selectors are server-side config and why the fallback is the default rather than the exception.

Q5.7 — "You call it a virtual business manager. GeM has no order API. So where do GeM orders appear?"

Think. There is a real hole here. docs/Master-Technical-Reference.md §9.3 tells us how to answer it, and the instruction is to not pretend.

Answer. They do not appear automatically. GeM orders land on the GeM seller dashboard and nowhere else.

How we handle it, in preference order:

1. **A cluster coordinator checks the GeM dashboard and marks the order in the admin console** (/gem-recon, admin page 5). **A real, fundable job — remote, over a phone call, never a field visit.**

2. **Email parsing**, if GeM sends order notification emails.
3. **Roadmap**: "GeM order API pending MoU."

Do not pretend we have GeM order sync. Stating the gap and showing the workaround reads as maturity. Faking it is a question we cannot survive.

What is automatic, so the manager claim is not empty:

Channel	Mechanism
Our marketplace	Direct DB write — instant, we own it
ONDC	Beckn callbacks (<code>on_confirm</code> , <code>on_status</code>) — true push
Amazon	Notifications API → queue; Orders API as fallback poll
Flipkart	Order Management Notification webhooks — confirmed to exist (closed §18 item 7)
GeM	[no] none — coordinator reconciles
Meesho	[no] none — manual

All of them normalise into **one canonical Order** with one state machine — placed → packed → shipped → delivered → settled — and one artisan inbox in their language with a voice notification. The unification is the product; the ingestion mechanism differs per channel and always will.

Q5.8 — "Two buyers on two platforms buy the same one-of-a-kind saree in the same second. What happens?"

Think. Do not claim to solve it. Nobody solves it. Say that, then show the mitigation that actually shrinks the problem, because that part is genuinely good.

Answer. One of them loses, and there is a window in which that is unavoidable. Propagation to Amazon and ONDC takes seconds to minutes. **That window cannot be closed. Every multichannel tool in the world lives with it.**

What we build (docs/Master-Technical-Reference.md §10.2):

- **Central inventory ledger** — single source of truth in our DB. **No channel is authoritative**
- **Atomic reserve-on-order with optimistic locking** — a `version` column; first write wins, second rejected
- **Immediate fan-out** to all channels on any change
- **Oversell resolution flow** — auto-cancel the losing order, apology plus priority-restock offer to that buyer, and **never let the artisan absorb the account-health hit silently**

The mitigation that actually shrinks the problem, and it is a product decision not a technical one: most handicrafts are replicable, not unique. A weaver can make another gamcha; a specific antique brass piece he cannot.

Type	Behaviour
Made-to-order (qty > 1 + lead time)	No race condition at all
Unique piece (qty = 1)	Reserve-and-race, small window

So we push artisans toward made-to-order wherever the craft allows — which is better for them anyway, because they never have to turn down demand. `is_made_to_order` and `lead_time_days` are in the schema from day one.

Breaks when. Genuinely unique pieces — antiques, one-off commissions, a single GI-tagged heirloom. There the window is real, the losing buyer is cancelled, and someone is disappointed. We chose to make that visible and compensated rather than invisible.